

ACA Medicaid Expansion and Healthcare Labor Markets: Wages and Employment Among Healthcare Providers in Expansion vs. Non-Expansion States

Abdallah Dalis*

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Abstract

The Affordable Care Act's Medicaid expansion delivered a large, staggered demand shock to the U.S. healthcare system, but how that shock reached the workers who deliver care is not well understood. Using American Community Survey microdata from 2009 to 2019 and the cross-state timing of expansion, I estimate an event-study difference-in-differences design for annual wages, hours worked per week, and employment probability across five occupational groups: nurses, physicians, specialists, support staff, and therapists. Healthcare providers absorbed the demand shock primarily through the intensive margin: hours worked rose 0.9 to 1.4 percent relative to non-expansion states and remained elevated, while employment probability stayed essentially flat. Wage gains were limited in the short run and concentrated among more specialized roles, with physicians capturing a 3.4 percent increase at the expansion year while nurses worked more hours without commensurate pay. The findings suggest that the financial benefits of coverage expansion were distributed unevenly across the workforce, with stronger-leverage occupations capturing wage gains and others absorbing the added workload. Pre-trends are not perfectly clean for nurses and therapists, so those subgroup estimates warrant caution.

Keywords: Medicaid expansion; Affordable Care Act; healthcare labor markets; difference-in-differences; event study; intensive margin; occupational heterogeneity.

JEL: I13, I18, J22, J31.

*DePaul University. Email: dalisabdallah@gmail.com. Prepared for ECO 518: Labor Economics. I thank Prof. Brian Phelan. All errors are my own.

Plain-Language Summary

The Affordable Care Act let states expand Medicaid, and most did, at different times. That gave millions more people insurance and sharply increased the demand for healthcare. This paper asks a question the headlines usually skip: when demand for care jumped, what happened to the people who actually do the work, the nurses, doctors, therapists, and support staff?

Comparing states that expanded against those that did not, and tracking each year around the change, a clear pattern emerges. The extra demand was met mostly by existing workers putting in *more hours*, not by hospitals hiring many more people. Roughly half an additional hour a week, about 26 hours a year, with little new hiring.

The raises did not follow the hours. Physicians, who have the most bargaining power, won the clearest pay increase. Nurses and support staff worked more without matching gains. So the money that flowed into the system did not reach the workforce evenly; it pooled where workers already had leverage.

One honest caveat: for a couple of groups, nurses and therapists, the data hint that they were already on a different path before expansion, so those specific results should be read with care. The bigger takeaway holds: expanding coverage helped patients and hospital finances, but whether it helped the workers depended a lot on which job they held.

1 INTRODUCTION

Enacted in 2010, the Affordable Care Act (ACA) represented a significant shift in the U.S. health-care system, expanding patient protection and health insurance coverage to nearly 44 million non-elderly low-income Americans. Two mechanisms in particular drove this unprecedented growth in coverage: the expansion of Medicaid eligibility to individuals with incomes up to 138% of the federal poverty level, and the implementation of a public health insurance marketplace. Before the ACA, Medicaid eligibility was limited to specific groups of low-income individuals, such as pregnant women, low-income parents, and people with disabilities. However, by 2014, the ACA's Medicaid expansion provision granted states the option to extend coverage to a broader population. As of 2026, 41 states, including the District of Columbia, have adopted Medicaid eligibility expansions, while 10 states have yet to do so (KFF 2025). This variation in staggered policy adoption across states has provided a unique opportunity for researchers to evaluate the impact of the ACA on a variety of outcomes.

The motivation of this research paper stems from the rising employment share of healthcare workers in the U.S. labor market. Even amid recessions and economic downturns, the healthcare sector has shown consistent employment growth, while other sectors often lag in comparison. One likely explanation is the inelastic demand for healthcare services and the stability it entails, making healthcare occupations an increasingly appealing choice for workers. However, despite the growing workforce, healthcare worker shortages have been a growing concern in the sector. The COVID-19 pandemic has further exacerbated these shortages, leading to increased workloads and burnout among healthcare workers (ASPE 2024). This apparent paradox of rising employment alongside frequent shortages highlights the importance of understanding the

dynamics of the healthcare labor market, particularly in response to policy changes that affect demand for healthcare services. Understanding the intensive and extensive margins across occupations can further inform policymakers about which roles are being relied on the most to meet the increased demand for healthcare services, and which roles may be more vulnerable to shortages.

Given our motivation, the primary focus of our research is on the healthcare labor market and its response to policy-induced demand shocks. For context, before the ACA, the uninsured rate for non-elderly adults was roughly 15 percent, whereas following the ACA, the uninsured rate fell to approximately 7.7 percent (KFF 2025). The largest increases in insurance coverage were a result of the Medicaid eligibility expansion, which led to roughly 22 million additional people gaining coverage, and an additional \$180 billion in state and federal Medicaid spending (KFF, 2025). As a result of this substantial rise in healthcare utilization and spending, the demand for healthcare workers increased significantly. Although the literature surrounding the ACA and its impact on insurance coverage, patient outcomes, and hospital outcomes is extensive, there is limited research pertaining to the ACA's impact on healthcare occupations and supply-side dynamics. Given this gap, our research seeks to find a causal impact of Medicaid expansion on healthcare labor market outcomes, including employment, wages, and hours worked. More specifically, we seek to identify how the effects differ across healthcare occupations, such as physicians, nurses, and support staff.

To evaluate this exogenous demand shock, we analyze American Community Survey (ACS) data from 2009 to 2019, matching state-year observations to state-level Medicaid expansion data collected from the Kaiser Family Foundation (KFF). For the empirical strategy, we utilize a difference-in-differences approach, where we exploit timing variation in the adoption of Medicaid expansion to identify a causal effect on healthcare labor market outcomes. The identifying assumption for our research is that expansion and non-expansion states would be similar in healthcare labor market trends, absent Medicaid expansion.

Overall, our findings suggest that healthcare providers responded to demand shocks primarily through intensive margins and lagged wage effects, with varying evidence regarding extensive margin shifts. Occupation-specific responses also varied, with positive wage effects being concentrated among more specialized fields, and intensive margins being more prevalent among roles with more flexible hours, such as nurses and support staff. We find some evidence of nurses and physicians seeing extensive margin responses; they appear to be modest and limited to specific time periods. These findings suggest that the financial benefits of coverage expansion were not distributed equally across the healthcare workforce: more specialized workers with stronger market leverage captured wage gains, while nurses and support staff absorbed the additional workload with limited compensation.

2 LITERATURE REVIEW

One of the biggest public health coverage pushes in American history came through the Affordable Care Act's extension of Medicaid, shifting how medical services are funded and affecting job needs across sectors (Courtemanche et al., 2017). Healthcare's role in driving new jobs gets frequent mention in mainstream reporting, especially where expanded eligibility led to stronger

staffing patterns and steadier clinic operations (DePillis & Zhang, 2025; Spetz et al., 2014). When hospitals closed, records show those in non-expansion regions faced higher risks, especially outside cities, because lost payments had strained budgets more severely (Daley, 2018). With broader enrollment, clinics saw income improve, not just from better patient payment profiles but also from fewer unpaid bills dragging on finances (Ammula & Guth, 2023). Meanwhile, broader access to care was likely to boost the need for medical staff, yet exactly how much, and where, stayed unclear. When combined, these perspectives cast Medicaid growth as a strong push on health systems, prompting curiosity about how more insured people lead to shifts in hiring patterns. Academic research provides evidence along several dimensions of this mechanism.

First, studies examining insurance coverage confirm that Medicaid expansion significantly increased enrollment in expansion states relative to non-expansion states (Courtemanche et al., 2017). Using difference-in-differences and triple-difference frameworks, this literature establishes that coverage gains were largest among low-income populations and in areas with high baseline uninsurance. These findings document the first-stage effect of expansion: increased insured demand for healthcare services, the demand-side mechanism that motivates our analysis.

Second, research directly examining provider labor markets finds evidence consistent with increased labor demand, though adjustments occur along multiple margins. Studies of the nurse labor market show that registered nurses and licensed practical nurses increased their hours worked and the probability of full-time employment in expansion states, particularly in rural regions and long-term care settings (DiNardi, 2021). Rather than entry into the profession driving these changes, existing workers adjusted along the intensive margin, suggesting stronger demand pressure on incumbent staff, a result we directly replicate and extend to a broader set of occupation groups. Related work examining newly trained primary care physicians finds that starting compensation rose more sharply in expansion states, especially for general internists and family physicians (Ma et al., 2022). Recent findings also show pay increases mainly went to better-compensated health workers, such as doctors and nurses, with little improvement seen among lower-earning aides (Matta et al., 2024). This uneven pattern suggests rising demand likely boosted wages mostly for those already holding stronger job leverage or distinct expertise, consistent with our finding that physicians captured disproportionately larger wage gains than nurses.

Even as projections highlight gaps in workforce capacity, evidence suggests that supply bottlenecks may dampen hiring even as patient demand climbs (Frogner et al., 2015). Credentialing barriers and long training pipelines constrain labor supply in healthcare in ways that have no analog in most other industries, which helps explain why employment may not expand even when demand increases sharply. Although analyses anticipated greater personnel requirements after policy changes, they rarely measured actual shifts in pay or job numbers (Frogner et al., 2015). On the financial side, hospital operating margins improved in Medicaid expansion states, particularly among safety-net hospitals serving large low-income populations (Ammula & Guth, 2023), and hospital closure rates fell (Daley, 2018), establishing that expansion increased revenues and financial stability, a necessary precondition for any downstream effect on worker compensation.

When pieced together, findings confirm that expanding Medicaid lifted enrollment rates, eased fiscal strain on clinics, and altered workloads and earnings for select clinical roles (Courtemanche et al., 2017; DiNardi, 2021; Ma et al., 2022). Still, most studies zoom in on single job types or isolated outcomes instead of mapping broader patterns in income and staffing across the sector. What remains less clear is how these impacts differ between high- and low-skilled positions, and how adjustment unfolds dynamically over time rather than in a single pre-post comparison. This paper contributes by examining wages, hours, employment, and annual earnings simultaneously across five occupational groups, using a fully specified event-study design that traces adjustment over five post-expansion years and explicitly tests the parallel trends assumption underlying the identification strategy.

3 DATA

This study utilizes microdata provided by the American Community Survey (ACS) to examine how Medicaid eligibility expansion under the Affordable Care Act affected labor market outcomes for healthcare workers. The ACS is an expansive survey, conducted annually by the U.S. Census Bureau, that collects cross-sectional data on individuals' demographic characteristics, including employment, income, and occupation status. Given the sheer count in the survey participation, it is widely recognized as a useful data source for studying time-varying labor market outcomes across states. For instance, the Census Bureau contacts over 3.5 million households annually to participate, and it is the largest household survey administered in the United States.

The analysis uses this ACS data covering the period 2009–2019, which allows for comparisons before and after the implementation of Medicaid expansion beginning in 2014. The sample is restricted to individuals ages 18–64 who are employed in healthcare occupations. Healthcare workers are identified using occupation codes 3000–3550, which include nurses, physicians, specialists, and healthcare support occupations. These restrictions ensure that the analysis will only focus on the working-age healthcare labor force who are most likely to be affected by the changes in the healthcare demand following the Medicaid expansion.

To evaluate differences between expansion adopters and non-expansion adopting states, states are classified into treated states (those that adopted Medicaid expansion) and control states (those that did not adopt expansion during the sample period). This classification allows for a difference-in-differences comparison of labor market outcomes with the two groups of states over a period. Furthermore, by restricting the sample to 2009–2019, we avoid confounding effects from the COVID-19 pandemic, which had a significant impact on healthcare labor markets starting in 2020. Looking at this timeframe also has the added benefit of increasing the pool of states in the control group, as several states adopted expansion in 2020 or later.

Table 1: Descriptive Statistics

	Treated states			Control states		
	Mean	SD	N	Mean	SD	N
Age	41.609	12.205	708382	41.125	12.106	426058
Female	0.780	0.414	708382	0.789	0.408	426058
White	0.738	0.440	708382	0.759	0.428	426058
Black	0.096	0.295	708382	0.149	0.356	426058
Asian	0.107	0.309	708382	0.052	0.221	426058
Other	0.059	0.235	708382	0.041	0.199	426058
BA or higher	0.514	0.500	708382	0.465	0.499	426058
Married	0.585	0.493	708382	0.593	0.491	426058

Descriptive statistics are computed from ACS microdata for 2009–2019.

The sample is restricted to healthcare occupations

The sample is further restricted to ages 18–64.

Table 1 reports summary statistics for the full sample. The average age of healthcare workers in this sample is approximately 41 years old, and at roughly 78 percent, the healthcare labor force shows a predominantly female population. The demographic composition of the workers in the expansion and non-expansion states is generally similar. For example, the average age and gender composition are nearly identical across the two groups of states. Education levels are also comparable, with roughly 51 percent of workers in expansion states and 47 percent in non-expansion states holding a bachelor’s degree or higher. These similarities suggest that the treated and control groups are broadly comparable in terms of observable demographic characteristics.

Table 2: Occupation Statistics

	Treated states			Control states		
	Mean	SD	N	Mean	SD	N
Nurse	0.361	0.480	708382	0.377	0.485	426058
Doctor	0.145	0.352	708382	0.127	0.333	426058
Specialist	0.237	0.425	708382	0.250	0.433	426058
Support Occup.	0.166	0.372	708382	0.164	0.370	426058
Therapist	0.091	0.288	708382	0.082	0.274	426058

Descriptive statistics are computed from ACS microdata for 2009–2019.

The sample is restricted to healthcare occupations

The sample is further restricted to ages 18–64.

Table 2 provides descriptive statistics on the occupational composition of the healthcare workers in this sample. Most workers fall into two main categories: specialists and nurses. Specialists, such as lab and machine technicians, represent roughly 24 percent of workers in expansion states and 25 percent in non-expansion states, while nurses account for about 36 percent and 37 percent of workers, respectively. Physicians represent about 15 percent of workers in expansion states and 13 percent in non-expansion states, while support staff account for roughly 16 percent for both groups. Finally, therapists account for about 9 percent and 8 percent of workers in expansion and non-expansion states, respectively. The occupational distribution is similar across the two groups of states, which again supports the idea that the treated and control groups are broadly comparable in terms of observable characteristics.

In addition to the demographic characteristics, we evaluate several labor market outcome trends for healthcare workers, including hourly wages, hours worked per week, and employment share in healthcare occupations. These figures are found in the appendix and allow us to assess how Medicaid expansions may have affected labor demand before our empirical analysis. Figure 4, for instance, shows the gap between occupational groups in terms of log wages, suggesting that physicians consistently earn more compared to other groups, while support staff earn the least. Note the relative stability in wages over time, as this will be relevant to our analysis later. In addition, Figure 5 shows the trends in hours worked per week, where we can see work hours becoming closer across occupational groups over time. Finally, as mentioned in the introduction, our motivation is derived from Figure 6, which shows us that healthcare workers are becoming a larger share of the overall labor market, motivating our research question about how the healthcare labor market is responding to demand shocks.

4 EMPIRICAL METHODOLOGY

The empirical strategy for this analysis is a difference-in-differences approach, which exploits the variation in the timing of Medicaid expansion across states to identify the causal effect of the policy on healthcare labor market outcomes. In the absence of the policy change, we assume that expansion and non-expansion states would have followed similar trends in healthcare labor market outcomes. This parallel trend assumption is needed for the identification of causal effects and will be further evaluated in the results section; however, we discuss potential violations and issues with our approach later in this section. For our methodology, we take inspiration from DiNardi (2021), who uses a similar approach to evaluate the impact of Medicaid expansion on the nurse labor market. We estimate the following regression for each labor market outcome of interest and for each healthcare occupation group, including nurses, physicians, support staff, specialists, and therapists.

$$Y_{ist} = \alpha + \sum_{k \neq -1} \beta_k D_{st}^k + X'_{it} \pi + \gamma_s + \delta_t + \varepsilon_{ist}. \quad (1)$$

In Equation (1), Y_{ist} represents the labor market outcome for an individual i in state s at time t . We examine three main outcomes: annual wage or salary income, hours worked per week, and the probability of employment in a healthcare occupation. The key coefficient that we will interpret is β_k , which captures the effect of Medicaid expansion at event time k relative to the year of expansion. This coefficient is attached to the indicator variable D_{st}^k , which equals 1 if the state s is k years away from expansion at time t , and 0 otherwise. Although represented in shorthand notation in the estimating equation, this term can also be portrayed as $D_{st} \times \mathbf{1}\{t - T_s = k\}$, where T_s is the year of expansion for state s . This indicator will turn on when an observation is from a state that expanded within our restricted period and give us the estimated effect of the policy at each event time relative to the control group. The event window we analyze spans 4 years before and 4 years after expansion, allowing us to trace the policy's effects over time and test for pre-trends in the years leading up to expansion. The omitted year is $k = -1$, which serves as the reference period for our estimated coefficients.

Following this is our vector of controls X_{it} , which includes demographic characteristics such as age, race, gender, education level, and marital status. In addition to these covariates, we control for the changes in healthcare demand by including a variable for the proportion of the population age 65 and older in state s at time t . We control for these variables to capture any confounding factors that may be correlated with both the outcome of interest and the policy change. For instance, if we assume that expansion states have a disproportionate number of elderly individuals, a group that typically consumes more healthcare services, the estimated effects of Medicaid expansion would be biased upward, as our β_k coefficient would likely capture both the policy-induced demand shock and the demand shock driven by demographic changes.

Finally, we control for state fixed effects γ_s and year fixed effects δ_t to account for unobserved heterogeneity across states and over time. Time-invariant differences across states, such as healthcare infrastructure, rural versus urban settings, and varying health policy, are captured by the state fixed effects. Year fixed effects control for shocks common to all states each year,

such as inflation, recessions, and other national-level events that may affect healthcare labor market outcomes outside Medicaid expansion. Capturing temporal and state trends is crucial for isolating the effect of Medicaid expansion on our outcomes of interest. Otherwise, our estimates may be biased, thereby limiting the validity of our findings. With this in mind, we turn to any potential concerns with our empirical approach. Although we control for a variety of confounding variables, several endogeneity concerns may affect the validity of our estimates. The largest concern with our approach is the obvious lack of random assignment into treated and control groups, as state self-selection into Medicaid expansion may introduce selection bias to our estimated effects. For instance, systematic differences between states, such as political ideology, may influence both healthcare labor market outcomes and the decision to expand Medicaid. We attempt to mitigate this concern by controlling for state fixed effects, relying on this method to capture all the relevant differences across states. This is a bold and difficult assumption to hold. Although having random assignment would be ideal, given the nature of the policy change and our dataset, it is unfortunately not possible. Furthermore, spillover effects and anticipation effects may also bias our estimates. For the latter, if healthcare providers anticipated the policy change, adjusting the labor supply of workers prior to the policy-induced demand shock, our estimating equation would produce biased estimates, and the parallel trends assumption would likely be violated. For instance, one could imagine a scenario where hospitals begin hiring more support staff in the year leading up to expansion. For the former, if newly insured patients in expansion states sought care in non-expansion states, this could lead to changes in labor market outcomes in the control group and underestimate the true effect of Medicaid expansion. We suspect that these effects are more pronounced in states that directly border expansion states, and a robustness check that excludes these states from the treatment group could prove useful in evaluating the extent of spillover effects. For the most part, Medicaid coverage is limited to in-state use, outside of emergency care. Some states have more lenient policies regarding counties that cross state lines, so this is a concern that should be evaluated further.

5 RESULTS

The results of our analysis are presented in the tables below. We begin by analyzing the effect of Medicaid expansion on the overall healthcare sector and then break down the effects by occupational group. Later in the section, we discuss the labor market dynamics associated with our findings, along with possible parallel trend violations. The coefficients are labeled by the event time relative to expansion, with $k = -1$ as the omitted reference period. We begin this section by analyzing the impact of Medicaid expansion on annual wage income.

Table 3 presents the estimated coefficients for the log of annual wage. For the healthcare sector, we see limited evidence of wage growth for the years immediately following expansion. In the long run, however, evidence suggests that healthcare workers experienced a 2.9 percent increase in annual wages relative to non-expansion states, on average. These results resemble an 8-lagged wage effect, which may be indicative of nominal wage rigidity and wage bargaining frictions in the healthcare labor market. However, to fully understand the dynamics of wage growth in the healthcare sector, we look to the other columns in Table 3, which break down the estimated wage effects by occupational group.

Notably, relative to non-expansion states, physicians saw the largest wage increase because of Medicaid expansion, with a 3.4 percent increase during the initial expansion period, and mostly growth in the years following, albeit statistically insignificant. Specialist occupations also appear to have a positive and increasing trend post-expansion; the statistical insignificance of each estimate must be noted. In contrast, support and therapist occupations experienced more volatility in wages, with trends leaning heavily towards negative growth in the years following the initial expansion period. For instance, relative to non-expansion states, support staff saw a 3.6 percent decline in annual wages 1-year post-expansion but later rebounded to a 2.6 percent increase 4 years post-expansion.

Finally, we found limited evidence of wage growth for nurses, with the estimated coefficients being largely insignificant. In addition, the parallel trends assumption for this group is questionable, as pre-trends suggest that expansion-state nurses were on different trajectories relative to non-expansion states. Overall, wage effects vary significantly across occupations, with evidence suggesting that more specialized roles, such as physicians and specialists, were able to capture greater wage effects from Medicaid expansion.

Table 3: Post ACA Expansion & Log Wage by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.010 (0.006)	-0.025** (0.009)	0.006 (0.017)	-0.021 (0.024)	0.005 (0.015)	-0.006 (0.024)
t-3	0.010 (0.007)	-0.011 (0.012)	0.005 (0.022)	0.002 (0.025)	0.023 (0.015)	0.029 (0.020)
t-2	0.003 (0.007)	-0.002 (0.010)	0.012 (0.019)	-0.029 (0.027)	0.008 (0.009)	-0.011 (0.021)
Expansion Year	0.006 (0.008)	0.005 (0.010)	0.034* (0.017)	-0.022 (0.023)	-0.007 (0.014)	0.006 (0.022)
t+1	0.007 (0.007)	0.010 (0.010)	0.017 (0.017)	-0.036* (0.018)	0.008 (0.014)	-0.019 (0.022)
t+2	0.005 (0.007)	0.009 (0.010)	-0.009 (0.024)	-0.015 (0.030)	0.015 (0.013)	-0.033* (0.017)
t+3	0.011 (0.008)	0.008 (0.009)	0.003 (0.020)	-0.012 (0.024)	0.021 (0.014)	-0.030 (0.018)
t+4	0.029*** (0.006)	0.009 (0.009)	0.031 (0.020)	0.026* (0.015)	0.021 (0.014)	0.031 (0.024)
Observations	1,035,427	396,391	150,077	147,379	248,498	93,082
R-squared	0.283	0.147	0.283	0.102	0.203	0.128

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4 presents the estimated coefficients on employment probability for all healthcare workers. For the healthcare labor market, we see little evidence of any change along the extensive margin, with our estimated coefficients being largely insignificant and close to zero. Despite this, by looking at the occupational breakdown, we see several estimates that stand out in both magnitude and significance. During the initial expansion period, relative to non-expansion

states, physicians saw a 0.6 percentage point increase in employment probability, while specialist occupations experienced a 0.7 percentage point decrease in employment probability. These trends, however, are limited to the initial expansion year. In the years following the initial expansion period, relative to non-expansion states, the employment probability for nurses increases by 0.6 percentage points, whereas the employment probability for therapists declines by 1 percentage point.

Altogether, evidence suggests varying occupation-specific employment responses to Medicaid expansion, with nurses and physicians seeing some positive effects along the extensive margin. The economic significance of these results should also be considered. For reference, a 0.6 percentage point increase in the probability of employment translates to roughly 6 additional healthcare workers per 1,000 people, which is arguably modest given the large increase in healthcare demand following Medicaid expansion. This suggests that labor supply did not fully adjust through new hiring alone, leading us to our next results.

Table 4: Post ACA Expansion & Employment by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.003 (0.002)	0.002 (0.003)	-0.003 (0.003)	-0.008 (0.009)	-0.004 (0.004)	-0.009** (0.004)
t-3	0.000 (0.002)	0.002 (0.003)	0.004 (0.003)	-0.006 (0.008)	-0.001 (0.005)	0.001 (0.005)
t-2	0.000 (0.002)	0.002 (0.002)	0.002 (0.003)	-0.001 (0.006)	0.000 (0.004)	-0.006 (0.005)
Expansion Year	0.000 (0.002)	0.003 (0.002)	0.006** (0.003)	0.000 (0.005)	-0.007* (0.004)	-0.003 (0.004)
t+1	0.002 (0.002)	0.005 (0.003)	0.001 (0.003)	0.003 (0.005)	-0.003 (0.005)	0.001 (0.006)
t+2	0.001 (0.001)	0.006** (0.002)	0.004 (0.002)	-0.005 (0.004)	-0.002 (0.003)	-0.012 (0.007)
t+3	-0.002 (0.002)	0.002 (0.003)	0.002 (0.003)	-0.004 (0.006)	-0.004 (0.005)	-0.010* (0.005)
t+4	0.003 (0.002)	0.003 (0.003)	0.002 (0.003)	0.004 (0.004)	0.002 (0.005)	-0.007 (0.005)
Observations	1,134,440	425,325	163,663	171,767	271,129	102,556
R-squared	0.022	0.034	0.019	0.010	0.014	0.024

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5 presents the estimated coefficients for log hours worked per week. For the healthcare sector, post expansion we see a consistent and significant positive trend relative to non-expansion states, suggesting that healthcare workers responded to the positive demand shock primarily through intensive margins, rather than extensive margins. Relative to no expansion states, we find significant evidence suggesting that hours worked per week increased by 0.9 to 1.4 percent in the years following expansion.

Breaking it down by occupation, we see that nurses responded immediately to the demand shock through intensive margins, with a 1.3 percent increase in hours worked per week. Following the initial expansion period, nurses consistently worked more hours per week, with increases ranging from 1 to 1.2 percent relative to non-expansion states. Specialists saw a similar, but larger effect 2 years post expansion, with evidence suggesting an increase in hours worked per week of about 1.1 to 1.9 percent, on average. Interestingly, support staff did not see any significant changes until 4 years post expansion, where we see a 1.6 percent increase in hours worked per week relative to non-expansion states.

Outside of these occupation groups, evidence of intensive margin responses is limited, with most estimates being statistically insignificant. Putting significance aside, we do find that therapist occupations experienced a positive trend in hours worked per week, ranging from a 0.9 to 1.9 percent increase relative to non-expansion states. Overall, these findings suggest that the demand shock from Medicaid expansion was primarily absorbed through intensive margin adjustments, with varying responses across occupations.

Table 5: Post ACA Expansion & Log Hours by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.000 (0.003)	-0.003 (0.004)	0.005 (0.006)	0.002 (0.008)	-0.002 (0.007)	-0.008 (0.008)
t-3	0.007** (0.003)	0.002 (0.005)	0.004 (0.006)	0.008 (0.013)	0.009 (0.007)	-0.000 (0.011)
t-2	0.001 (0.003)	0.006 (0.004)	0.004 (0.006)	-0.013 (0.013)	-0.009 (0.006)	-0.000 (0.007)
Expansion Year	0.010*** (0.004)	0.013** (0.005)	0.012* (0.007)	0.009 (0.009)	0.004 (0.007)	0.012 (0.010)
t+1	0.009** (0.004)	0.010** (0.005)	0.002 (0.009)	0.002 (0.009)	0.005 (0.008)	0.019 (0.012)
t+2	0.009*** (0.003)	0.012** (0.005)	0.007 (0.007)	0.000 (0.008)	0.011* (0.006)	0.009 (0.012)
t+3	0.009** (0.003)	0.010** (0.004)	-0.002 (0.007)	-0.000 (0.011)	0.015* (0.008)	0.008 (0.009)
t+4	0.014*** (0.003)	0.011* (0.006)	0.001 (0.007)	0.016** (0.007)	0.019** (0.007)	0.016 (0.010)
Observations	1,064,424	398,975	159,443	155,427	253,127	97,452
R-squared	0.054	0.026	0.074	0.029	0.050	0.040

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Altogether, our results suggest that healthcare providers, specifically nurses, support staff, and specialists, responded to the Medicaid expansion demand shock primarily through intensive margins, rather than extensive margins, with varying evidence of wage growth. Specifically, we found that nurses showed the most compelling evidence regarding intensive margin responses, suggesting that they were more likely to adjust their hours in response to the demand shock. There is some evidence of nurses and physicians also seeing extensive margin responses,

however they appear to be modest and limited to specific time periods. We also find evidence suggesting that wage growth is concentrated among specialized fields, such as physicians and specialists, while nurses and support staff saw limited evidence of any increase. These findings are relatively consistent with the results of previous literature, specifically DiNardi (2021) and Matta et al. (2024), with some notable differences regarding the impacts on nurse wages.

Importantly, when evaluating these results, we must consider the parallel trends assumption and whether the assumption sufficiently holds. For the most part pre-trends remained statistically insignificant, however some areas raise valid concerns regarding causality. For instance, annual wages for nurses in expansion states appear to be on different trends relative to non-expansion states, especially in the fourth year leading up to expansion. Similarly, we see differences in pre-trend for the employment probability among therapists as well. Although these concerns do not necessarily invalidate our findings, they do suggest that the estimated effects for these specific groups should be interpreted with caution. These divergences in pre-trends may be indicative of both the non-random assignment of states into treated and control groups, as well as the presence of anticipation effects, which we discussed in the empirical methodology section.

The most surprising takeaway from our results is the potential for increased hours per week without corresponding annual wage increases. Perhaps this lack of compensation, despite greater work intensity, may be indicative of the rising burnout and job dissatisfaction among healthcare occupations. Note that our estimates for hours per week translates to roughly half an hour in additional work, adding up to roughly 26 additional hours per year. Although modest in magnitude, uncompensated increases in hours worked can have significant implications for worker well-being, especially when considering the already demanding nature of healthcare jobs. In line with the findings of Frogner et al. (2015), our results for extensive margin responses support the idea that supply bottlenecks may have dampened hiring even as patient demand climbed, which may have led to workers absorbing additional workloads through intensive margin adjustments.

Finally, relating our findings to a broader economic framework, the demand shock from Medicaid expansion likely increased the marginal product of labor for healthcare workers, which theoretically should have led to increases in both wages and employment. However, the presence of wage rigidity and bargaining frictions in the healthcare labor market may explain the opposing wage effects we find in our results, as it could take time for wages to fully adjust. Additionally, the varying responses across occupations may be explained by differences in labor supply elasticity. For instance, although healthcare occupations are generally considered to be inelastic, physicians are typically more inelastic relative to other healthcare workers, which could explain why physicians were able to capture wage gains, while nurses, specialists, and support staff absorbed the additional workload.

It is also important to consider the typical work weeks each occupation encounters, as physicians typically have more rigid work schedules and longer shifts, while nurses and support occupations are typically more flexible in scheduling and hours, which could explain why the latter groups were more likely to adjust along the intensive margin. Overall, our findings suggest that the financial benefits of coverage expansion were not distributed equally across the healthcare sector, with inherent occupational differences affecting how providers responded to the demand shock from Medicaid expansion.

6 CONCLUSION

This paper examines the causal effect of ACA Medicaid expansion on labor market outcomes for healthcare workers using an event-study difference-in-differences design with American Community Survey microdata from 2009 to 2019. Exploiting cross-state variation in the timing of expansion, we estimate the effect of the policy on annual wages, hours worked per week, and employment probability across five occupational groups: nurses, physicians, specialists, support staff, and therapists.

Our central finding is that healthcare providers responded to the Medicaid expansion demand shock primarily through the intensive margin. Hours worked per week increased significantly beginning in the expansion year and remained elevated throughout the post-expansion window, with the overall sample showing increases ranging from 0.9 to 1.4 percent relative to non-expansion states. Employment probability, by contrast, remained largely flat across all occupational groups, suggesting that labor supply did not adjust through new hiring to any meaningful degree. This pattern is consistent with credentialing barriers and long training pipelines constraining the extensive margin of supply in ways that leave incumbent workers absorbing additional workloads instead.

Wage effects were limited in the short run and concentrated among specific occupational groups in the long run. For the overall healthcare workforce, annual wages showed no significant response until four years post-expansion, when we estimate a 2.9 percent increase relative to non-expansion states. Physicians were the notable exception, with a 3.4 percent wage gain at the expansion year itself, consistent with their stronger bargaining position and lower labor supply elasticity, translating demand gains into immediate compensation. Nurses, by contrast, saw consistent and significant hours increases throughout the post expansion period with no corresponding wage gains at any point in the event window. This divergence between nurses and physicians is the most striking finding of our analysis. It suggests that the financial benefits of coverage expansion were not distributed equally across the healthcare workforce: more specialized workers with stronger market leverage captured wage gains, while nurses and support staff absorbed the additional workload without commensurate compensation.

These findings speak directly to broader debates about the welfare consequences of health insurance expansions. Medicaid expansion clearly benefited patients and improved hospital finances, as the prior literature documents. Whether those gains flowed through to the workers who deliver care depends heavily on occupation. Policymakers interested in improving healthcare worker well-being may need to complement insurance expansions with targeted interventions, such as nurse staffing ratios, wage floors, or enhanced collective bargaining rights, to ensure that demand-driven gains are distributed more broadly across the workforce. The uncompensated hours increase documented here, which translates to roughly 26 additional work hours per year, carry real implications for the burnout and turnover pressures already straining the sector.

This study has several limitations that point toward productive directions for future research. First, the parallel trends assumption is not perfectly clean for all subgroups, particularly nurses, where the pre-expansion wage trend raises some concern, and results for that group should be interpreted with appropriate caution. Second, our sample is restricted to 2019 to avoid COVID-era confounding, which limits our ability to speak to longer-run dynamics. Third, extending

the analysis to states that expanded after 2019 and incorporating job mobility as an additional outcome, such as rate of hires and quits, would provide a more complete picture of the welfare effects for workers across the full distribution. Further research should also explore how effects vary by healthcare setting, rather than just occupation, as the demand shock may have played out differently in hospitals versus outpatient clinics. Along those same lines, rural vs urban differences are also worth exploring.

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A APPENDIX: SUPPORTING FIGURES

Figure 1: Healthcare Worker Employment Share Over Time

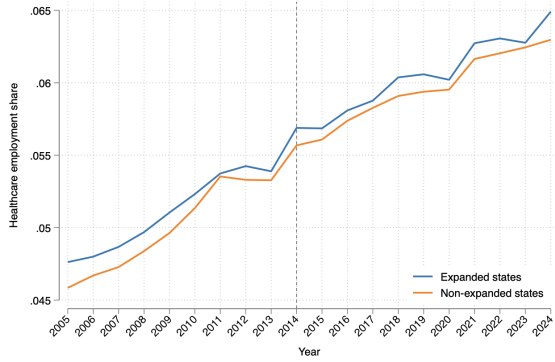


Figure 2: Healthcare Worker Log Wage Over Time

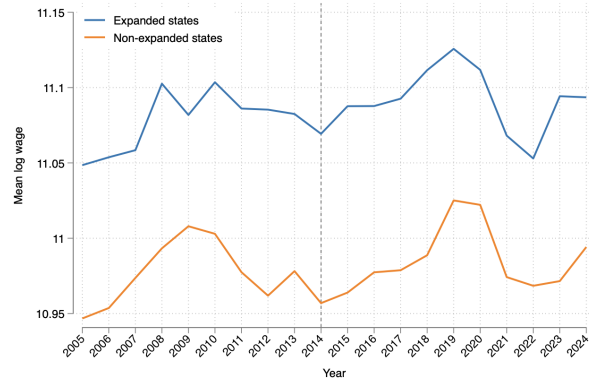


Figure 3: Healthcare Worker Hours/Week Over Time

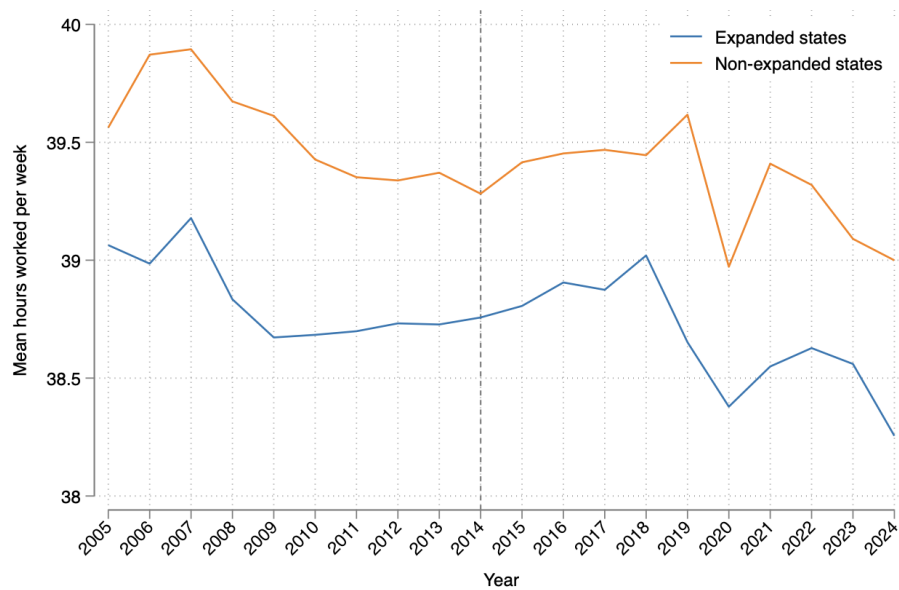


Figure 4: Healthcare Worker Log Wage Over Time by Occupation

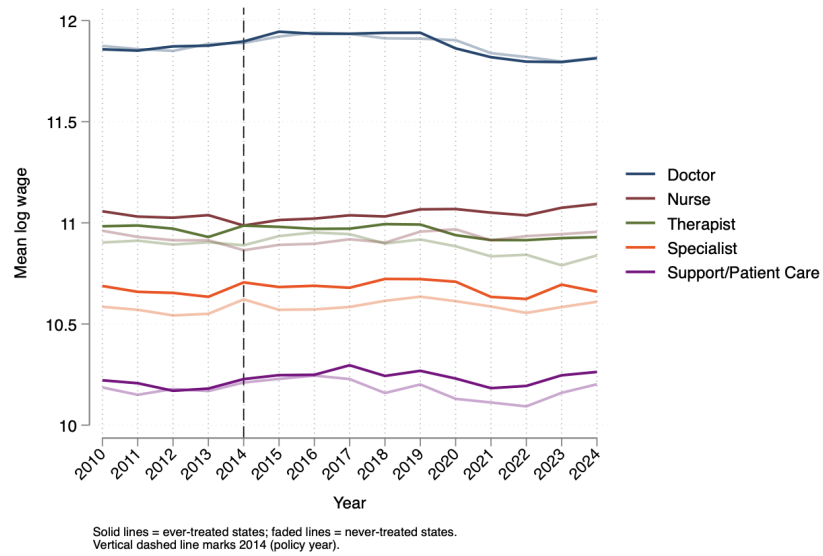


Figure 5: Healthcare Worker Hours/Week Over Time by Occupation

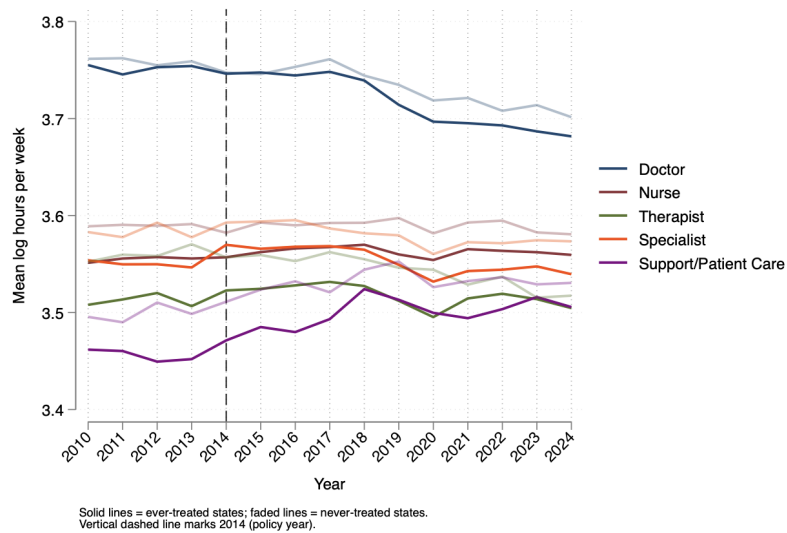


Figure 6: Event Study Coefficients for Log Annual Wage

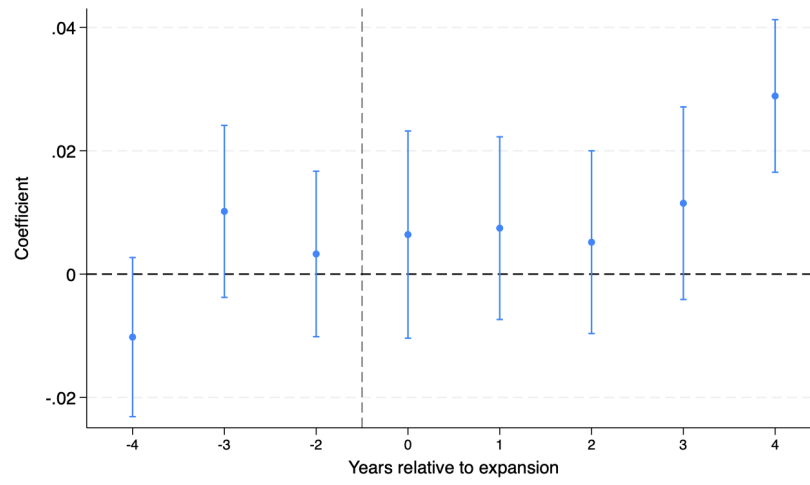


Figure 7: Event Study Coefficients for Log Annual Employment

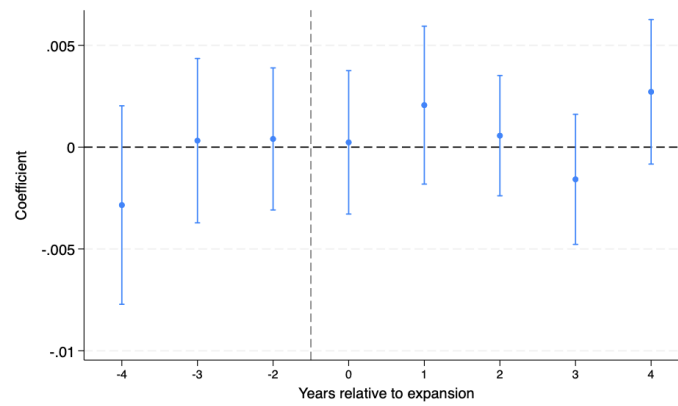


Figure 8: Event Study Coefficients for Log Hours/Week

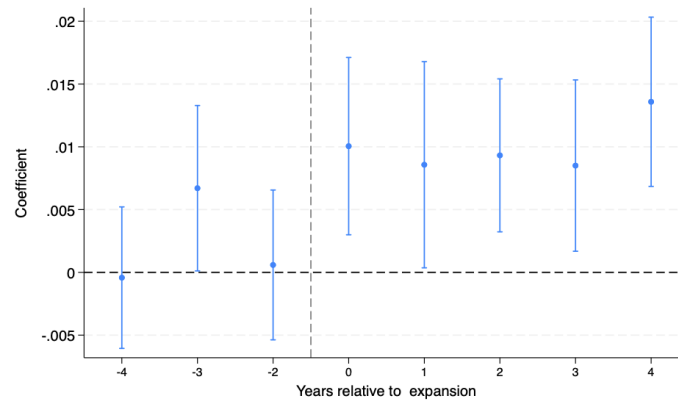


Table 6: Descriptive Statistics

	Treated states			Control states		
	Mean	SD	N	Mean	SD	N
Age	41.609	12.205	708382	41.125	12.106	426058
Female	0.780	0.414	708382	0.789	0.408	426058
White	0.738	0.440	708382	0.759	0.428	426058
Black	0.096	0.295	708382	0.149	0.356	426058
Asian	0.107	0.309	708382	0.052	0.221	426058
Other	0.059	0.235	708382	0.041	0.199	426058
BA or higher	0.514	0.500	708382	0.465	0.499	426058
Married	0.585	0.493	708382	0.593	0.491	426058

Descriptive statistics are computed from ACS microdata for 2009–2019.

The sample is restricted to healthcare occupations

The sample is further restricted to ages 18–64.

Table 7: Occupation Statistics

	Treated states			Control states		
	Mean	SD	N	Mean	SD	N
Nurse	0.361	0.480	708382	0.377	0.485	426058
Doctor	0.145	0.352	708382	0.127	0.333	426058
Specialist	0.237	0.425	708382	0.250	0.433	426058
Support Occup.	0.166	0.372	708382	0.164	0.370	426058
Therapist	0.091	0.288	708382	0.082	0.274	426058

Descriptive statistics are computed from ACS microdata for 2009–2019.

The sample is restricted to healthcare occupations

The sample is further restricted to ages 18–64.

Table 8: Post ACA Expansion & Log Wage by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.010 (0.006)	-0.025** (0.009)	0.006 (0.017)	-0.021 (0.024)	0.005 (0.015)	-0.006 (0.024)
t-3	0.010 (0.007)	-0.011 (0.012)	0.005 (0.022)	0.002 (0.025)	0.023 (0.015)	0.029 (0.020)
t-2	0.003 (0.007)	-0.002 (0.010)	0.012 (0.019)	-0.029 (0.027)	0.008 (0.009)	-0.011 (0.021)
Expansion Year	0.006 (0.008)	0.005 (0.010)	0.034* (0.017)	-0.022 (0.023)	-0.007 (0.014)	0.006 (0.022)
t+1	0.007 (0.007)	0.010 (0.010)	0.017 (0.017)	-0.036* (0.018)	0.008 (0.014)	-0.019 (0.022)
t+2	0.005 (0.007)	0.009 (0.010)	-0.009 (0.024)	-0.015 (0.030)	0.015 (0.013)	-0.033* (0.017)
t+3	0.011 (0.008)	0.008 (0.009)	0.003 (0.020)	-0.012 (0.024)	0.021 (0.014)	-0.030 (0.018)
t+4	0.029*** (0.006)	0.009 (0.009)	0.031 (0.020)	0.026* (0.015)	0.021 (0.014)	0.031 (0.024)
Observations	1,035,427	396,391	150,077	147,379	248,498	93,082
R-squared	0.283	0.147	0.283	0.102	0.203	0.128

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Post ACA Expansion & Employment by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.003 (0.002)	0.002 (0.003)	-0.003 (0.003)	-0.008 (0.009)	-0.004 (0.004)	-0.009** (0.004)
t-3	0.000 (0.002)	0.002 (0.003)	0.004 (0.003)	-0.006 (0.008)	-0.001 (0.005)	0.001 (0.005)
t-2	0.000 (0.002)	0.002 (0.002)	0.002 (0.003)	-0.001 (0.006)	0.000 (0.004)	-0.006 (0.005)
Expansion Year	0.000 (0.002)	0.003 (0.002)	0.006** (0.003)	0.000 (0.005)	-0.007* (0.004)	-0.003 (0.004)
t+1	0.002 (0.002)	0.005 (0.003)	0.001 (0.003)	0.003 (0.005)	-0.003 (0.005)	0.001 (0.006)
t+2	0.001 (0.001)	0.006** (0.002)	0.004 (0.002)	-0.005 (0.004)	-0.002 (0.003)	-0.012 (0.007)
t+3	-0.002 (0.002)	0.002 (0.003)	0.002 (0.003)	-0.004 (0.006)	-0.004 (0.005)	-0.010* (0.005)
t+4	0.003 (0.002)	0.003 (0.003)	0.002 (0.003)	0.004 (0.004)	0.002 (0.005)	-0.007 (0.005)
Observations	1,134,440	425,325	163,663	171,767	271,129	102,556
R-squared	0.022	0.034	0.019	0.010	0.014	0.024

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Post ACA Expansion & Log Hours by Occupation

	All HCWs	Nurses	Doctors	Support	Specialists	Therapists
t-4	-0.000 (0.003)	-0.003 (0.004)	0.005 (0.006)	0.002 (0.008)	-0.002 (0.007)	-0.008 (0.008)
t-3	0.007** (0.003)	0.002 (0.005)	0.004 (0.006)	0.008 (0.013)	0.009 (0.007)	-0.000 (0.011)
t-2	0.001 (0.003)	0.006 (0.004)	0.004 (0.006)	-0.013 (0.013)	-0.009 (0.006)	-0.000 (0.007)
Expansion Year	0.010*** (0.004)	0.013** (0.005)	0.012* (0.007)	0.009 (0.009)	0.004 (0.007)	0.012 (0.010)
t+1	0.009** (0.004)	0.010** (0.005)	0.002 (0.009)	0.002 (0.009)	0.005 (0.008)	0.019 (0.012)
t+2	0.009*** (0.003)	0.012** (0.005)	0.007 (0.007)	0.000 (0.008)	0.011* (0.006)	0.009 (0.012)
t+3	0.009** (0.003)	0.010** (0.004)	-0.002 (0.007)	-0.000 (0.011)	0.015* (0.008)	0.008 (0.009)
t+4	0.014*** (0.003)	0.011* (0.006)	0.001 (0.007)	0.016** (0.007)	0.019** (0.007)	0.016 (0.010)
Observations	1,064,424	398,975	159,443	155,427	253,127	97,452
R-squared	0.054	0.026	0.074	0.029	0.050	0.040

Standard errors in parentheses

Standard errors clustered at the state level. ACS weights used (pwgtp).

All models include state and year fixed effects and controls.

Sample restricted to employed workers, 2009–2019.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$